Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

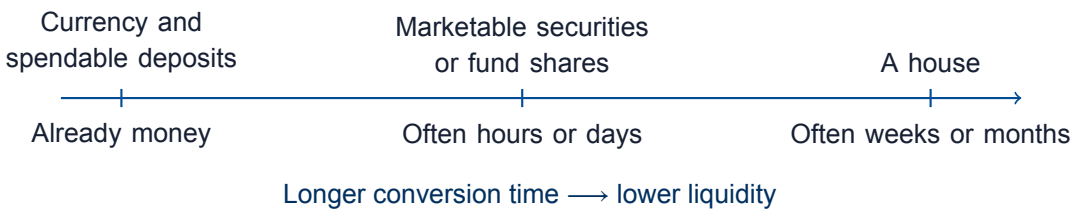
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Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

Money and Liquidity

- **Money**: assets that can readily be used to purchase goods and services.
- **Liquidity**: how quickly an asset can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money
Pay with a portion of the crop accepted as money. It also has value as food.

Commodity-backed money
Pay with a warehouse receipt redeemable for a specified quantity of grain.

Fiat money
Pay with officially accepted money that promises no commodity redemption.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or invest it in other assets.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

T-Accounts

A **T-account** lists **assets on the left** and **liabilities on the right** to summarize a bank or business.

Assets		Liabilities	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Assets: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' equity.

Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Bank Runs

- A **bank run** occurs when many depositors withdraw because they fear the bank may fail.
- The bank holds only part of its funding as immediately available reserves; loans cannot all be collected at once.
- Forced asset sales can cause losses, increasing fear and triggering more withdrawals.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher rates reduced long-term securities' values; technology-sector customers drew down deposits.
- **March 8–9, 2023**: SVB announced securities losses and a capital plan; the next day, deposits fell by more than $40 billion.
- **March 10, 2023**: California closed the bank. Many large, uninsured and connected depositors made the run especially fast.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Regulation

- **Deposit insurance**: FDIC created in 1933; coverage began in 1934. Eligible deposits: up to $250,000 per depositor, insured bank, and ownership category.
- **Discount window**: collateralized Fed loans help banks meet temporary funding needs.
- **Interest on reserve balances (IORB)**: the Fed pays interest on eligible reserve balances and helps guide short-term rates.
- **Capital requirements**: a minimum cushion of owners' equity relative to exposures absorbs losses.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.
- Deposit insurance can weaken depositors' incentive to monitor a bank. Owners may gain from risky bets while others bear part of the losses.
- **Capital requirements** put more of owners' funds at risk; supervision complements that incentive.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: credit intermediation outside ordinary deposit banking.
- Examples: **investment banks**, **money market funds**, hedge or credit funds, finance companies, and securitization vehicles.
- Some insurance-group activities also belong to this chain. Short-term funding of illiquid assets can create run-like risks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Shadow Banking in the Great Recession

Date	Institution	What happened
March 2008	Bear Stearns Investment bank	Funding run; acquired with Fed support.
September 15, 2008	Lehman Brothers Investment bank	Filed for bankruptcy.
September 16, 2008	AIG Insurance group	Collateral pressure; emergency Fed support.
September 16, 2008	Reserve Primary Fund Money market fund	Lehman losses; share value fell below $1.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Source: Federal Reserve Board, AIG assistance, September 16, 2008.

Money and Bank Claims

- A deposit is money to its holder and a liability of the bank.
- Reserves support payments; equity absorbs losses.
- Part 2: the Fed, loan creation, settlement, repayment, and deposit expansion.

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Chapter 14, Part 2

Money Creation and the Federal Reserve

Guilherme Paiva Pinto

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Roadmap

1. The Federal Reserve System
2. Loan creation, settlement, and repayment
3. A conditional deposit-expansion model
4. Practice and an individual checkpoint

The Federal Reserve System

Institution	Role
Board of Governors	Federal oversight and system responsibilities
12 Reserve Banks	Regional operations, financial services, and supervision
Federal Open Market Committee (FOMC)	Decisions about the monetary-policy stance

1913: the Federal Reserve Act established the Fed to address banking-system stresses.

Source: Federal Reserve Board, The Fed Explained.

Bank Accounting

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

Reserves and loans are assets. Deposits are liabilities. Equity is the residual.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Reserves remain 200; equity remains 100.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Bank B: reserves +120; seller’s deposits +120.

The Banking System

Change since before the loan	Bank A	Bank B	Combined
Reserves	−120	+120	0
Loans	+120	0	+120
Deposits	0	+120	+120

The payment moves deposits; the loan created the additional $120 of deposits.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.
- Public currency rises by $50, offsetting the fall in deposits.
- Currency + deposits is unchanged; the bank has less liquidity.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.
- Loans fall by $50 and deposits fall by $50.
- Currency + deposits falls by $50; reserves and equity do not change.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loans fall to 820; equity falls to 20.

Deposits and reserves remain unchanged at the write-off.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.
- **Risk and profitability**: borrowers must qualify, and lending must cover costs.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.
- All loan proceeds are redeposited; no extra reserves or currency leakage.
- Banks lend each available amount; borrowers want the loans; other constraints do not bind.

Successive Deposit Rounds

Round	New deposit ($)	Reserves retained	New loan
1	100.00	10.00	90.00
2	90.00	9.00	81.00
3	81.00	8.10	72.90

The central bank first buys a $100 security from a nonbank seller, adding reserves and a deposit.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.
- These are limiting model results under the stated conditions.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

With the same expansion conditions, deposits rise by 1,000 in total, but currency + deposits rises by 900.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.
- $1/r$ at $r = 0$ does not predict infinite lending or an infinite money supply.

Source: Federal Reserve Board, Reserve Requirements.

Central-Bank Security Purchases

Transaction	Reserves	Nonbank deposits
Buy from a bank	Bank reserves rise	No immediate required change
Buy from a nonbank seller	Seller’s bank reserves rise	Seller’s deposit rises

Both transactions exchange assets; the immediate effect depends on the seller.

Balance-Sheet Practice

Use the worksheet's bank: reserves 150, loans 750, securities 100, deposits 900, equity 100.

Record a new $90 loan, then its payment to another bank.

For each event, verify assets = liabilities + equity and state what happens to total deposits.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

Loans −30 and equity −30.

Deposits and reserves do not change solely because of the write-off.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.
- The multiplier depends on its model conditions.
- Chapter 15 connects central-bank policy, interest rates, spending, and AD.

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.